

Interpretation of results:

BLASTn:

In the nucleotide sequence we used (*HBB* gene), query coverage ranged from 90% to 100% in all 20 results. Most of the sequence had 100% query coverage which mean there is little variation among the nucleotide sequences portions that matched and almost all region are conserved. Percentage identity remained above 95%, the lowest percentage being 96.50%, indicating that among the portion that matched, the number of nucleotides that aligned are almost perfect.

BLASTp:

In the protein sequence we used (HBB protein), query coverage ranged from 87% to 100%. One result had 87%, two had 99% and the rest (17) had 100% query coverage, which means most of the protein portion, if not all of the protein, aligned and had conserved domains. As for the percentage identity, only 1 result was 100% similar in the protein portion tat was conserved, but otherwise, percentages ranged from 83.67% to 93.20% which shows that there are some differences in the aligned portions, but the sequences still have a close evolutionary relationship.